

PART-A

Name of the Student: Raj Nandani Kumari
Signature of the Student: *Raj Nandani* Roll No.: 230123048
Signature of the Invigilator: *Ganbar m. meher*

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space **will not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	2	8	15

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Design matrix :

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$$

We know that

$$Y = X\theta + \epsilon$$

$$\hat{Y} = X\hat{\theta}, \hat{\theta} \text{ is the LSE of } \theta$$

where $\hat{\theta}$ is given by

$$\hat{\theta} = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 0.2222 & 0.1111 \\ 0.1111 & 0.2222 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \begin{bmatrix} 0.3333 & 0 & 0.3333 \\ 0.3333 & -0.3333 & 0 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \begin{bmatrix} 0.3333 & 0 & 0.3333 \\ 0.3333 & -0.3333 & 0 \end{bmatrix}$$

Now, $\hat{\theta} = (X^T X)^{-1} X^T y$

Therefore, the LSE of θ_1 and θ_2 can be written as

$$\hat{\theta}_1 = 0.3333 y_1 + 0.3333 y_3$$

$$\hat{\theta}_2 = 0.3333 y_1 - 0.3333 y_2$$

} Ans.

05

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \quad \begin{aligned} X_1 &= U_1 \\ X_2 &= U_2 \\ X_3 &= U_1 + U_2 \end{aligned}$$

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

Since $U_1 \sim U(0,1)$
 $U_2 \sim U(0,1)$

Therefore $\text{Var}(U_1) = \frac{(b-a)^2}{12} = \frac{1}{12} = \text{Var}(X_1) = \sigma_{11}$

$\text{Var}(U_2) = \frac{(b-a)^2}{12} = \frac{1}{12} = \text{Var}(X_2) = \sigma_{22}$

$\text{Var}(X_3) = \text{Var}(U_1 + U_2)$
 $= \text{Var}(U_1) + \text{Var}(U_2) + \text{Cov}(U_1, U_2)$
 $= \frac{1}{12} + \frac{1}{12} + \text{Cov}(U_1, U_2)$

$\text{Cov}(X_1, X_3) = \text{Cov}(U_1, U_1 + U_2)$
 $= \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2)$
 $= \text{Var}(U_1) + \text{Cov}(U_1, U_2)$
 $= \frac{1}{12} + \text{Cov}(U_1, U_2)$

Let $\text{Cov}(U_1, U_2)$ be c

Therefore, Σ becomes -

$$\Sigma = \begin{bmatrix} 0.5 & 0 & 0.5+c \\ c & 0.5 & 0.5+c \\ 0.5+c & 0.5+c & 1+c \end{bmatrix}$$

$\text{Cov}(U_1, U_2) = 0$ since both are independent uniform random variables on $(0,1)$.

$$X = [X_1, X_2, X_3]^T, \quad X_1 = U_1, \quad X_2 = U_2, \quad X_3 = U_1 + U_2$$

$$\text{Var}(U_1) = \frac{(b-a)^2}{12} = 0.5 \quad \text{Var}(U_2) = 0.5$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var } X} \sqrt{\text{Var } Y}}$$

Since U_1 and U_2 are independent R.V on $(0,1)$ $\therefore \text{Cov}(U_1, U_2) = 0$

$$\text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2) = 0.5 + 0.5 + 0 = 1$$

Therefore, correlation matrix becomes -

$$\rho = \begin{bmatrix} 1 & 0 & 0.7 \\ 0 & 1 & 0.7 \\ 0.7 & 0.7 & 1 \end{bmatrix}$$

$$\text{Also, } \text{Cov}(U_1 + U_2, U_1) = \text{Cov}(U_1, U_1) + \text{Cov}(U_2, U_1) = 0.5 + 0 = 0.5$$

$$\rho = \frac{\text{Cov}(U_1 + U_2, U_1)}{\sqrt{\text{Var}(U_1 + U_2)} \sqrt{\text{Var } U_1}} = \frac{0.5}{\sqrt{0.5} \sqrt{0.5}} = 0.7$$

Now, let us find the variance of first principal component -
we will first find the eigen values of ρ ($\lambda_1, \lambda_2, \dots$)

Then, $\lambda_1 > \lambda_2 > \dots$
 λ_1 will become the variance of first principal component
($\rho - \lambda I$) = 0

$$\begin{vmatrix} 0.5-\lambda & -\lambda & 0.7-\lambda \\ -\lambda & 0.5-\lambda & 0.7-\lambda \\ 0.7-\lambda & 0.7-\lambda & 1-\lambda \end{vmatrix} = 0$$

$$\lambda = \frac{1.15}{0.24} = 4.79$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

i) Complete linkage

Initially, we have

$\{1\}, \{2\}, \{3\}, \{4\}$

	1	2	3	4
1	0	1	11	5
2	1	0	2	3
3	11	2	0	3.5
4	5	3	3.5	0

Smallest distance = 1

So, we will cluster 1 and 2 together.

Now, we have $\{12\}, \{3\}, \{4\}$

	12	3	4
12	0	11	5
3	11	0	3.5
4	5	3.5	0

$$d(12, 3) = \max(d(1, 3), d(2, 3)) = \max(11, 2) = 11$$

$$d(12, 4) = \max(d(1, 4), d(2, 4)) = \max(5, 3) = 5$$

Now, the smallest distance is 3.5, so will cluster 3 and 4 together $\{12\}, \{34\}$

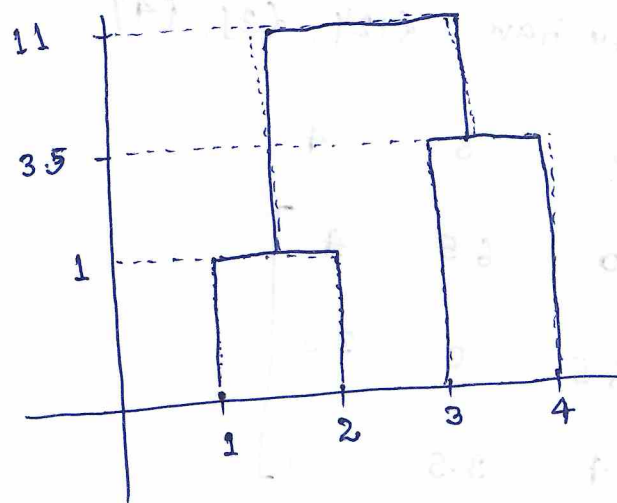
$$\begin{array}{cc} 12 & 34 \\ \begin{array}{c} 12 \\ 34 \end{array} & \begin{bmatrix} 0 & 11 \\ 11 & 0 \end{bmatrix} \end{array}$$

$$d(12, 34) = \max(d(12, 3), d(12, 4))$$

$$= \max(11, 5)$$

Now, cluster becomes $\{1234\}$ with distance 11

Dendrogram for complete linkage -



ADDITIONAL PAGES FOR ANSWERS

ii) Now, let us do the same for average linkage

Initial clusters = $\{1\}, \{2\}, \{3\}, \{4\}$

	1	2	3	4
1	0	1	11	5
2	1	0	2	3
3	11	2	0	3.5
4	5	3	3.5	0

Smallest distance = 1
 We will cluster 1 and 2 together
 Now, we have $\{12\}, \{3\}, \{4\}$

	12	3	4
12	0	6.5	4
3	6.5	0	3.5
4	4	3.5	0

$$d(12,3) = \frac{d(1,3) + d(2,3)}{2} = \frac{11+2}{2} = 6.5$$

$$d(12,4) = \frac{d(1,4) + d(2,4)}{2} = \frac{5+3}{2} = 4$$

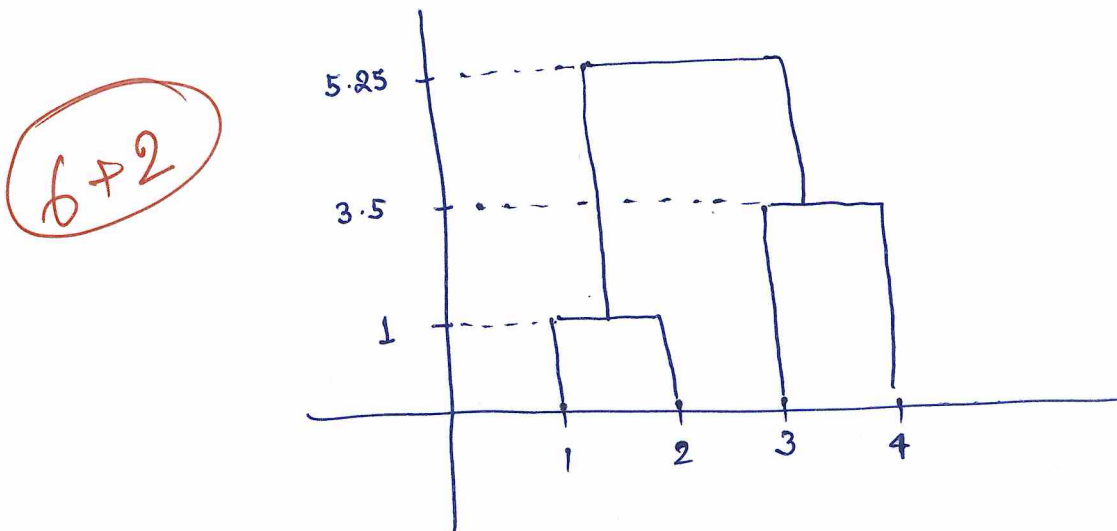
Now, smallest distance becomes 3.5
 So clusters are $\{12\}, \{34\}$

$$\begin{array}{cc}
 12 & 34 \\
 12 & \begin{bmatrix} 0 & 5.25 \\ 5.25 & 0 \end{bmatrix} \\
 34 &
 \end{array}$$

$$\begin{aligned}
 d(12,34) &= \frac{d(12,3) + d(12,4)}{2} \\
 &= \frac{6.5 + 4}{2} \\
 &= \frac{10.5}{2} = 5.25
 \end{aligned}$$

Now, the final cluster becomes $\{1234\}$ with distance b/w $\{12\}$ and $\{34\}$ be 5.25.

Dendrogram for average linkage:



Comments ??

